

DECISION TREE INDUCTION

ATTRIBUTE SELECTION MEASURES

31139 - Samit Kadam
31140 - Arjun Kankal
31180 - Pradyum Umale



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INTRODUCTION : DECISION TREE

- A decision tree is a flowchart-like tree structure, where each internal node denotes a test on an attribute, each branch represents an outcome of the test, and each leaf node holds a class label.
- Given a record/data, X , for which the associated class label is unknown, the attribute values of the tuple are tested against the decision tree. A path is traced from the root to a leaf node, which holds the class prediction for that tuple.
- The construction of decision tree classifiers does not require any domain knowledge or parameter setting, and therefore is appropriate for exploratory knowledge discovery.
- Decision trees can handle multidimensional data easily and pretty fast.
- Application areas are such as medicine, financial analysis (eligibility for loans), molecular biology, etc.

INTRODUCTION : ATTRIBUTE SELECTION MEASURES

- It is a metric for selecting the “splitting criterion” that “best” separates a given labelled data, containing of class-labelled training tuples, into individual classes.
- Ideally each partition would be pure, realistically it approaches the ideal situation, but never meets it.
- If the splitting attribute is continuous-valued, then a split point must also be determined as part of the splitting criterion.

INFORMATION GAIN

- It is used to find the minimum information that is needed to classify the records in the resulting partitions.
- It minimizes the expected number of tests needed to classify a given record and guarantees that a simple (but not necessarily the simplest) tree is found.
- Information contained in dataset D is given as:

$$Info(D) = - \sum_{i=1}^n \{p_i * \log_2(p_i)\}$$

where,

p_i : is the nonzero probability that an arbitrary record in D belongs to class C_i , given

by $\frac{\text{Number of records of class } C_i}{\text{Total number of records}}$

INFORMATION GAIN

- Now, suppose we were to partition the records in D on some attribute A , which may result splitting D into v partitions or subsets, $\{D_1, D_2, \dots, D_v\}$. Then Information Gain by attribute A is:

$$Info_A(D) = \sum_{i=1}^v \left\{ \frac{\text{total number of records in subset } D_i (|D_i|)}{\text{total number of records in } D (|D|)} * Info(D_i) \right\}$$

- Information gained by branching on A is given as:

$$Gain(A) = Info(D) - Info_A(D)$$

- The attribute A with the highest information gain, $Gain(A)$, is chosen as the “Splitting Attribute”.
- If A comprises of discrete values, then we will map each value to a single, unique partition (one – to – one mapping).
- If A comprises of continuous values, then we will define “Split points”, one partition in between each such point.

GAIN RATIO

- The information gain measure is biased toward tests that returns many outcomes/subsets.
- For e.g. : suppose attribute R is a unique identifier for the records. The information gain upon spitting on R will always be maximum, as it will result in pure classification of records. But such type of spitting is useless for classification.
- It measure the spitting by information gained and weights of corresponding subsets.
- A new parameter $SplitInfo_A(D)$ is defined as:

$$SplitInfo_A(D) = - \sum_{i=1}^v \left\{ \frac{|D_i|}{|D|} * \log_2 \left(\frac{|D_i|}{|D|} \right) \right\}$$

GAIN RATIO

- $SplitInfo_A(D)$ represents the potential information generated by splitting the training data set, D, into v partitions, corresponding to the v outcomes of a test on attribute A.
- Finally, Gain Ratio is defined as:

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

- As $SplitInfo_A(D)$ approaches 0(zero) , the $GainRatio$ becomes unstable and unreliable. Hence, extra constraint is added, which is : the information gain of the test selected must be **large**—at least as great as the average gain over all tests examined.

GINI INDEX

- Unlike others, Gini Index measures the spilt of dataset D into 2 parts, D_1 and D_2 .
- It measures the impurity of D.
- Given by formula:

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

- It assumes dataset division into 2 parts. These two parts may contain even overlapping classes/labels.

GINI INDEX

- Assuming attribute A be the split criterion, if it contains v discrete values, then there are total $2^v - 2$ possible ways to two partition set. Gini index for choosing A as split criterion is given by:

$$Gini_A(D) = \frac{|D_1|}{|D|} Gini(D_1) + \frac{|D_2|}{|D|} Gini(D_2)$$

- The reduction in impurity that would be incurred by a binary split on attribute A is given by:

$$\Delta Gini(A) = Gini(D) - Gini_A(D)$$

EXAMPLE AND APPLICATION : ABOUT DATASETS

A. Iris dataset:

1. Number of attributes: 4
2. About attributes: length and width of both sepal and petal; all continuous attributes
3. Dimensions: 150 records x 5 attributes (including class)
4. Class: Setosa, Versicolor, Virginica

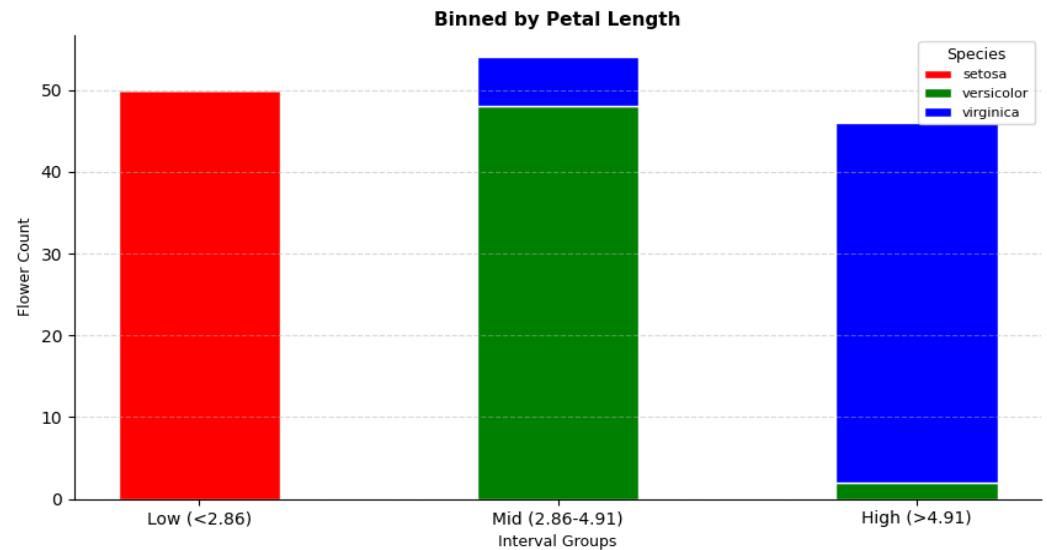
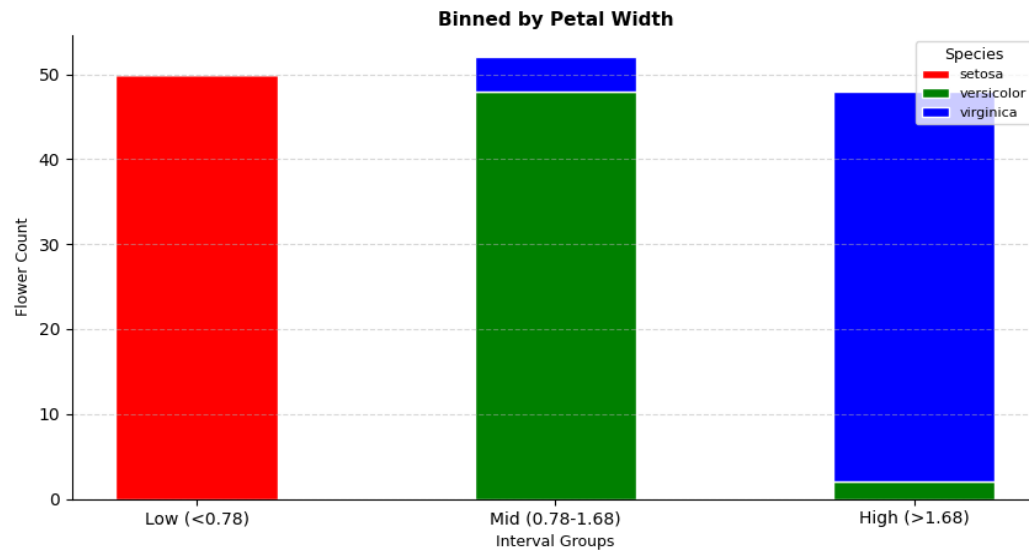
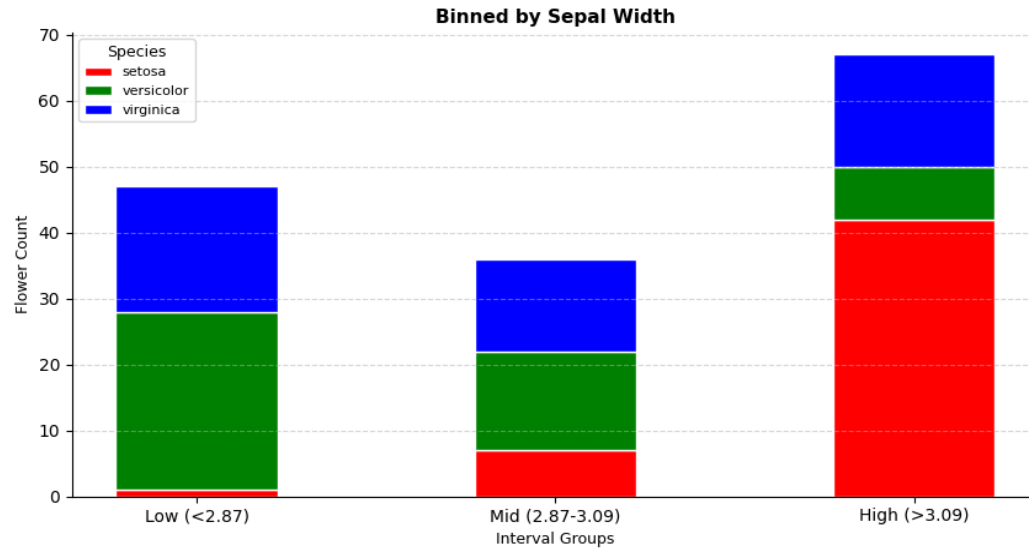
B. Lung cancer dataset:

1. Number of attributes: 15
2. About attributes: gender, age and other health factors; all binary, except age(continuous)
3. Dimensions: 309 records x 16 attributes (including class)
4. Class: YES, NO

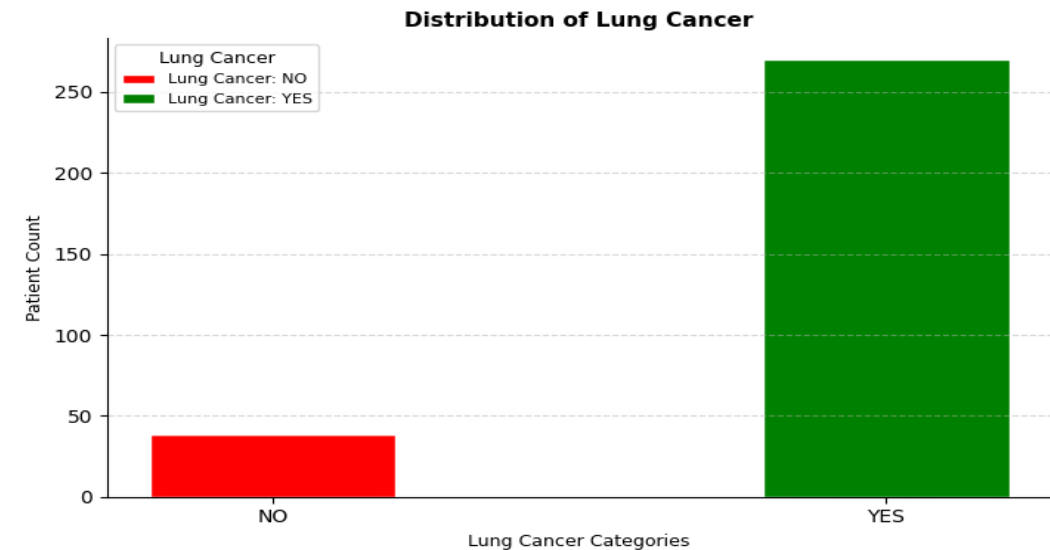
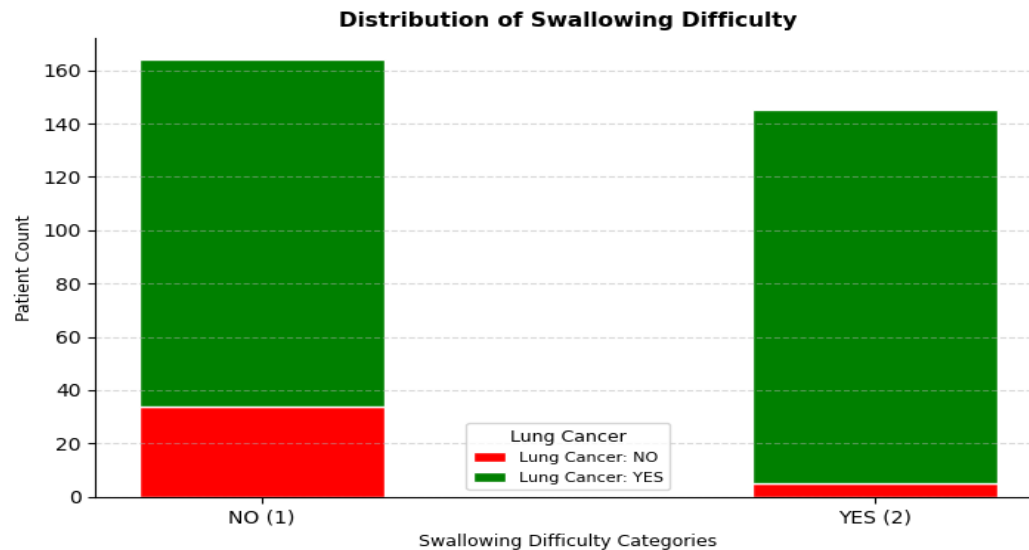
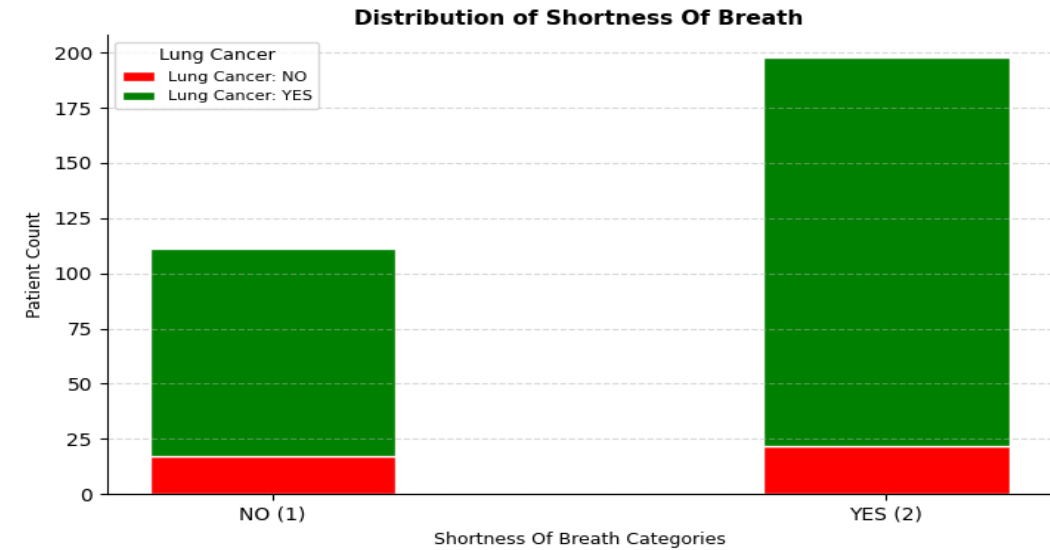
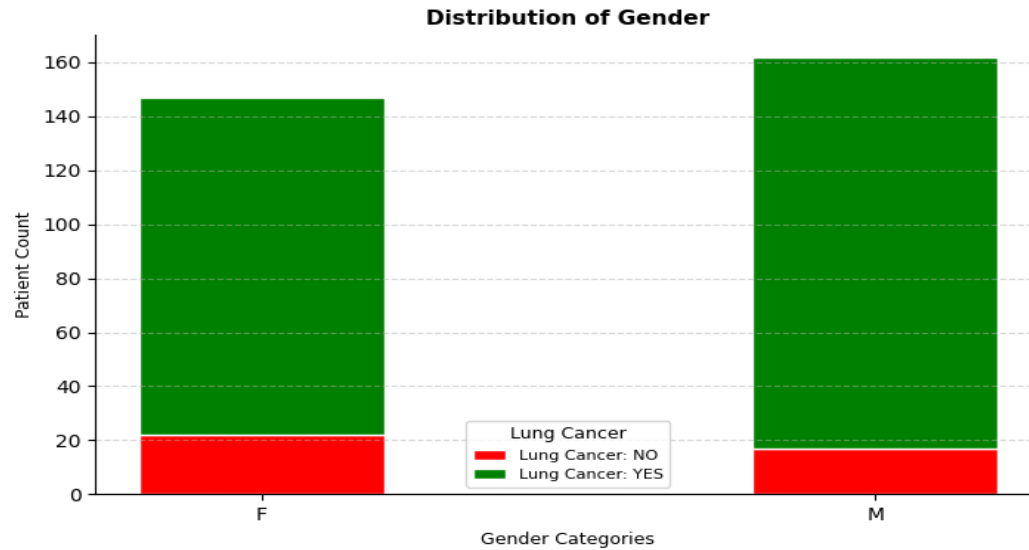
C. Credit Approval dataset:

1. Number of attributes: 15
2. About attributes: mixture of 6 continuous and 9 categorical data (total 42 categories)
3. Dimensions: 653 records x 16 attributes (including class)
4. Class: +, -

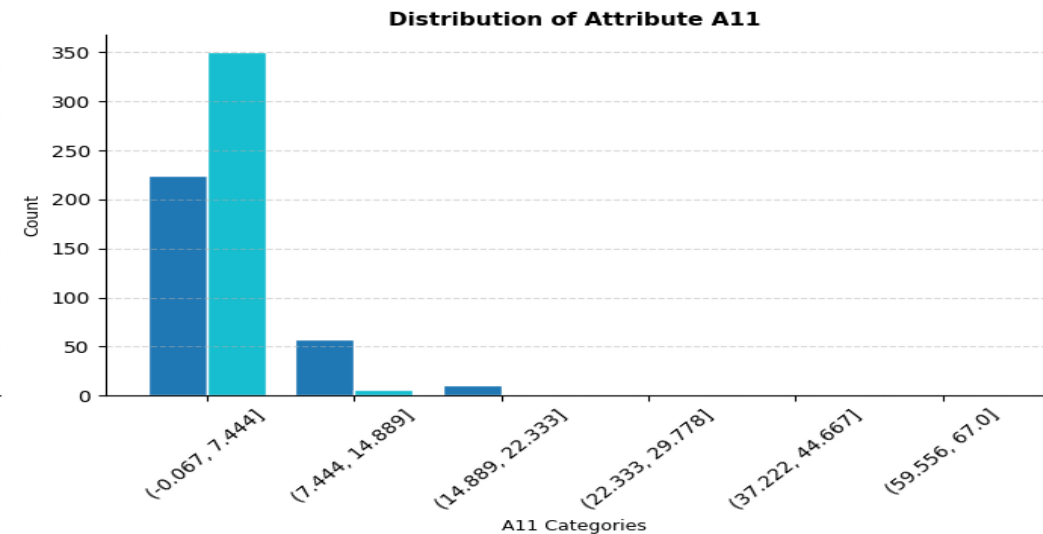
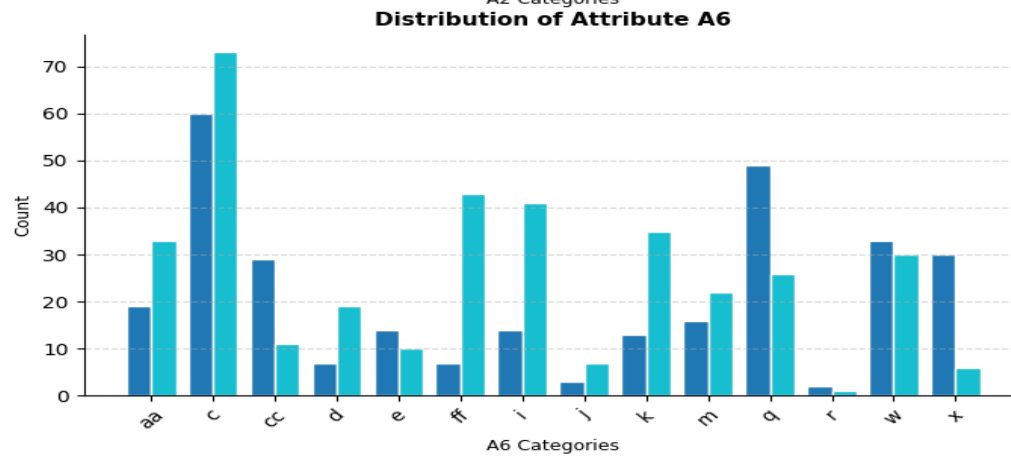
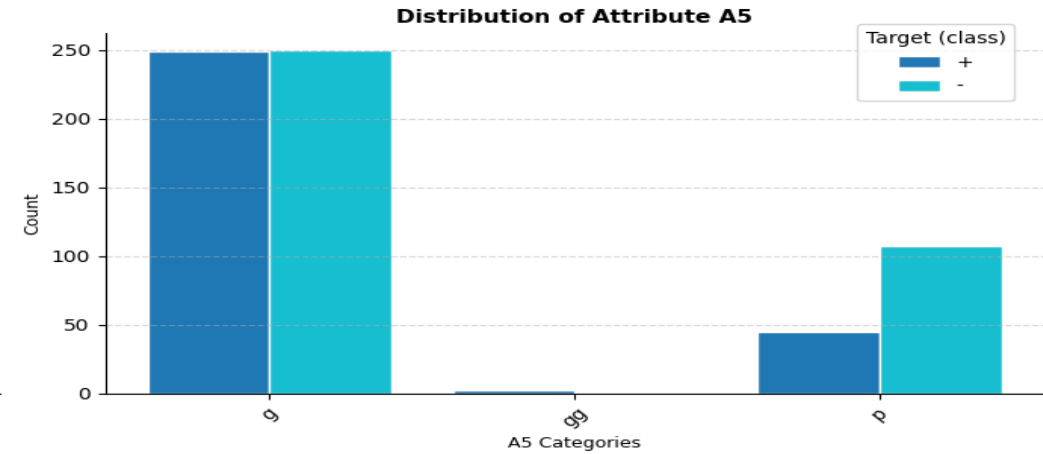
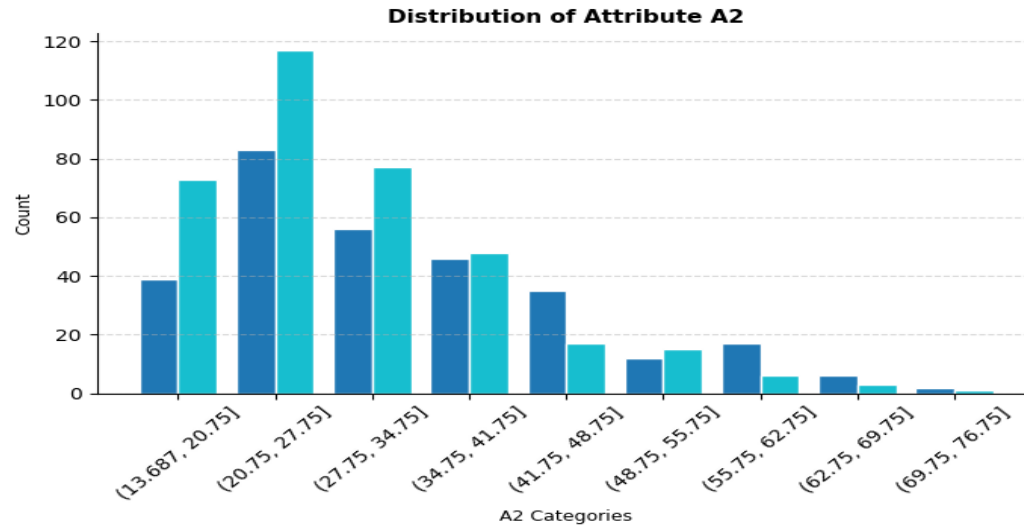
EXAMPLE AND APPLICATION : IRIS DATASET GRAPHS



EXAMPLE AND APPLICATION : CANCER DATASET GRAPHS



EXAMPLE AND APPLICATION : CREDIT DATASET GRAPHS



EXAMPLE AND APPLICATION : INFORMATION GAIN

Python 3.13 (64-bit)

```
>>> exec(open('info_gain.py').read())
```

```
Iris : Info(D) = 1.5850
```

	sepal_width	sepal_length	petal_width	petal_length
Info(A,D)	1.2866	0.9083	0.2156	0.2603
Info Gain(A)	0.2983	0.6766	1.3694	1.3247

```
Lung Cancer : Info(D) = 0.5470
```

	GENDER	SHORTNESS OF BREATH	SWALLOWING DIFFICULTY	LUNG_CANCER
Info(A,D)	0.5437	0.5444	0.4923	0.000
Info Gain(A)	0.0033	0.0026	0.0546	0.547

```
Credit Approval : Info(D) = 0.9937
```

	A2	A5	A6	A11
Info(A,D)	0.9800	0.9682	0.8887	0.8003
Info Gain(A)	0.0137	0.0255	0.1050	0.1934

```
>>>
```

EXAMPLE AND APPLICATION : GAIN RATIO

```
Windows PowerShell
>>> exec(open("split_info.py").read())
Iris : Info(D) = 1.5850
      sepal_width  sepal_length  petal_width  petal_length
Info(A,D)         1.2866        0.9083        0.2156        0.2603
Info Gain(A)       0.2983        0.6766        1.3694        1.3247
SplitInfo(A)       1.5381        1.5836        1.5842        1.5819
Gain Ratio(A)      0.1940        0.4273        0.8644        0.8374

Lung Cancer : Info(D) = 0.5470
      GENDER  SHORTNESS OF BREATH  SWALLOWING  DIFFICULTY  LUNG_CANCER
Info(A,D)    0.5437                0.5444        0.4923        0.000
Info Gain(A) 0.0033                0.0026        0.0546        0.547
SplitInfo(A) 0.9983                0.9420        0.9973        0.547
Gain Ratio(A) 0.0033                0.0028        0.0548        1.000

Credit Approval (CRX) : Info(D) = 0.9937
      A2      A5      A6      A11
Info(A,D)  0.9800  0.9682  0.8887  0.8003
Info Gain(A) 0.0137 0.0255 0.1050 0.1934
SplitInfo(A) 0.9775 0.8116 3.5081 0.8408
Gain Ratio(A) 0.0140 0.0315 0.0299 0.2300
>>> |
```

EXAMPLE AND APPLICATION : GINI INDEX

```
Python 3.13 (64-bit) x + v
>>> exec(open('gini_index.py').read())
Iris : Gini(D) = 0.6667
      sepal_width sepal_length petal_width petal_length
Gini_A(D)      0.5476      0.3794      0.0748      0.0966
Reduction      0.1191      0.2872      0.5919      0.5700

Lung Cancer : Gini(D) = 0.2206
      GENDER SHORTNESS OF BREATH SWALLOWING DIFFICULTY LUNG_CANCER
Gini_A(D)  0.2196      0.2198      0.2057      0.0000
Reduction  0.0010      0.0008      0.0149      0.2206

Credit Approval (CRX) : Gini(D) = 0.4956
Attribute division:
A5: (['g', 'p'], ['gg'])
A6: (['aa', 'c', 'd', 'ff', 'i', 'j', 'k', 'm'], ['cc', 'e', 'q', 'r', 'w', 'x'])
      A2      A5      A6      A11
Gini_A(D)  0.4862  0.4938  0.4497  0.3714
Reduction  0.0094  0.0018  0.0459  0.1243
>>> |
```

EXAMPLE AND APPLICATION : GINI INDEX

```
Python 3.13 (64-bit) × + v
>>> exec(open('gini_index.py').read())
Iris : Gini(D) = 0.6667
      sepal_width sepal_length petal_width petal_length
Gini_A(D)      0.5476      0.3794      0.0748      0.0966
Reduction      0.1191      0.2872      0.5919      0.5700

Lung Cancer : Gini(D) = 0.2206
      GENDER SHORTNESS OF BREATH SWALLOWING DIFFICULTY LUNG_CANCER
Gini_A(D)  0.2196      0.2198      0.2057      0.0000
Reduction  0.0010      0.0008      0.0149      0.2206

Credit Approval (CRX) : Gini(D) = 0.4956
Attribute division:
A5: (['g', 'p'], ['gg'])
A6: (['aa', 'c', 'ff', 'cc', 'e', 'q', 'r'], ['d', 'i', 'j', 'k', 'm', 'w', 'x'])
      A2      A5      A6      A11
Gini_A(D)  0.4862  0.4938  0.4940  0.3714
Reduction  0.0094  0.0018  0.0016  0.1243
>>> |
```

REFERENCES

1. Book: Data Mining: Concepts and Techniques 3rd Edition Morgan Kaufmann 2011
2. Iris dataset: <https://www.kaggle.com/datasets/himanshunakrani/iris-dataset>
3. Lung cancer dataset: <https://www.kaggle.com/datasets/mysarahmadbhat/lung-cancer>
4. Credit Approval dataset: <https://www.kaggle.com/datasets/echo9k/uci-credit-approval/data>
5. Sample code: https://github.com/PradyumUmale/dwm_presentation_grp4



THANK You